Predoc Data Task

Submission

Please submit:

- A single PDF document containing your written responses, figures, and tables, and
- All relevant code files (e.g., Stata .do file, R script, Python script, etc.) necessary to fully reproduce your results.

You have 48 hours to complete the task.

Part 1: Paid Family Leave

Context:

The United States is the only OECD country lacking a national Paid Family Leave (PFL) program. In the absence of a federal policy, several states have implemented paid family leave programs. One of the earliest was New Jersey's Family Leave Insurance (FLI) program, enacted in 2009, which provides eligible workers with up to six weeks of paid leave at a 66% wage replacement rate.

You will evaluate whether the introduction of New Jersey's FLI program in 2009 improved the economic well-being of households most likely to be affected. New Jersey shares a labor market and economic environment with neighboring Pennsylvania within the Philadelphia Metropolitan Area, yet Pennsylvania did not implement a paid family leave policy during this period.

Eligibility for FLI is determined at the **household level** and depends on (i) sufficient labor force attachment of a designated adult household member (the "reference individual"), defined as **having worked at least 20 weeks in the prior year**, and (ii) the presence of a qualifying caregiving need **(e.g., a recent birth, a child less than one years old in the household, or caring for another household member with a disability**).

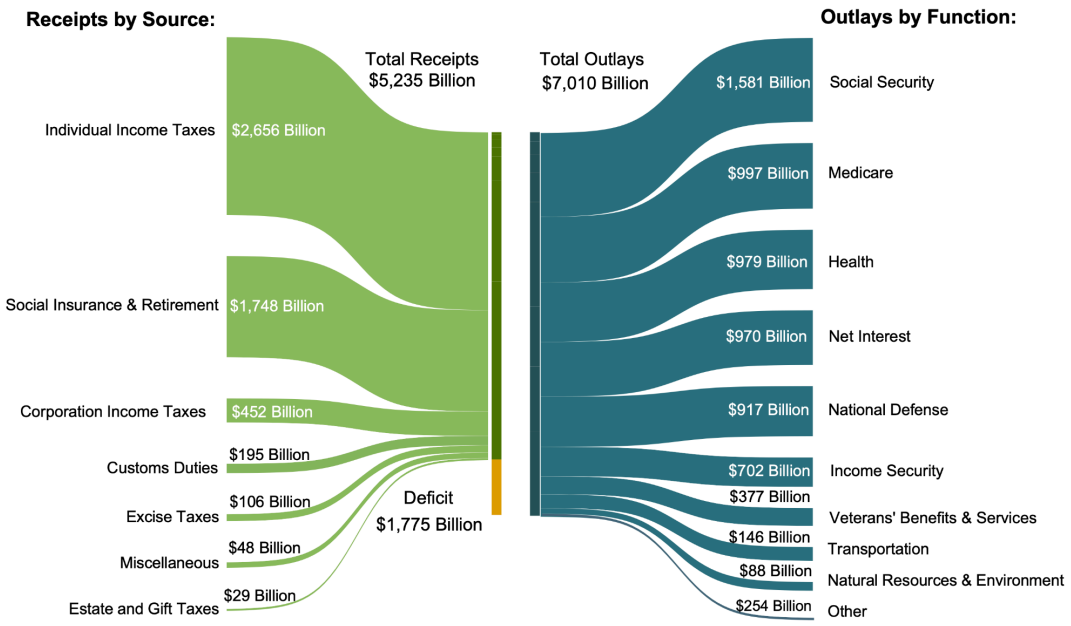
You are provided with individual-level cross-sectional American Community Survey (ACS) 1-Year PUMS data (via IPUMS USA) for households in the Philadelphia Metropolitan Area from 2006–2012. A codebook describing all variables is included.

Because the ACS is collected annually, variable definitions and coding may change over time.

1. Import the ACS PUMS data and perform any necessary cleaning and preprocessing steps.
2. Tabulate and summarize the distribution of weeks worked in the prior year by state and year. Comment on whether/how the distribution changes over time.
3. Construct a binary indicator identifying households in both states that plausibly meet the eligibility criteria for New Jersey's FLI policy. Use this indicator to create a publication-ready graphic showing how mean household income evolves over time for eligible v.s. non-eligible households, separately for New Jersey and Pennsylvania.
    a. Based on your visualization, can the figure be interpreted as evidence of causal effects of New Jersey's FLI policy? Why or why not? Discuss.
4. Please estimate the effect of New Jersey's FLI policy on household income by running two regressions: one using household income in levels and one using the natural log of household income. Present the results in two columns, and for each specification, report the regression equation corresponding to the regression you run. Which specification do you think is more appropriate in this setting—levels or logs? Explain your reasoning.
5. Being eligible for New Jersey's FLI program does not necessarily mean a household *takes* the leave to which they're entitled. How does incomplete take-up affect the interpretation of estimates that use eligibility as the treatment variable? What is an empirical strategy that could help address this issue if additional data were available, and clearly state the assumptions required for this strategy to identify a causal effect.

Part 2: Sankey Diagram Exercise. Understanding State Budget Inflows and Outflows

The figure below presents a Sankey Diagram for the US Federal Budget in 2024, published by the US Treasury in reports available here.



Task:

We would like you to produce an analogous Sankey diagram representing the budget of the state of Pennsylvania.

Specifically, please construct a Sankey diagram that illustrates the flow of Pennsylvania's state budget from revenue sources to expenditure categories for a recent fiscal year. You may use publicly available Pennsylvania state budget documents, revenue reports, or summary tables as your data sources. Your diagram should not separately display every line item of receipts and outlays. Rather, like the example above, it should aggregate them into sensible and informatively named categories, such as personal income tax, sales tax, corporate tax (for revenues) and education, health and social services, transportation (for outlays).

To produce the diagram, you may also use any tools you like to produce this graphic, including dedicated software tools, online data visualization tools, or AI tools.

In your submission document, provide the Sankey diagram and explain how you constructed it. Specifically, please explain how you obtained the data you used, what steps you took to clean, aggregate, or modify that data,, and how you constructed the visualization itself, referencing any external tools you used. Focus on your reasoning and design choices, and attach any code you used.